EXPERIMENT 13

Implementation of a Ping Program

EXP:13
Date: 26/7/26

PING program

AIM:

To implement a simple ping program in computer networks to check whether a destination host is reachable and to measure the round-trip of packets

Algorithm:

1) Start the program
2) Read the destination host name or IP address from the user
3) create a raw ICMP socket
4) calculate a ICMP Echo Request Packet containing:
   - ICMP Type
   - code
   - che ck sum
   - Data
5) Calculate the checksum of the packet
6) Send the ICMP Echo Request to the destination
7) Display the final result and remove

User-agents

March 2004 | Last 12 Months | Monthly Statistics | Daily Statistics | Hourly Statistics | URLs

Entry Pages | Exit Pages | Errors | Hosts | Referrer | Searches | Usernames | Useragents | Countries

9	72	4.66%	71	5.57%	52	5.30%	86	0.80%	71	30.34%	0.09	6.62	inktomisearch.com
10	4	0.26%	4	0.31%	4	0.41%	74	0.69%	4	1.71%	0.00	0.00	rhyolite.com

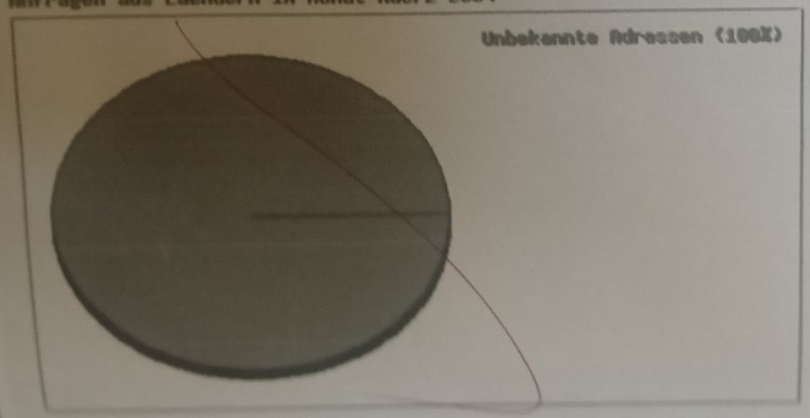
Top 1 von 1 Verweise

#	Anfragen		Besuche		Verweise
1	1546	100.00%	234	100.00%	- (Direkte Anfrage)

Top 1 von 1 Anwenderprogramme

#	Anfragen		kb		Besuche		Anwenderprogramm
1	1546	100.00%	10699	100.00%	234	100.00%	-

Anfragen aus Laendern in Monat Maerz 2004



Top 1 von 1 Ländern

#	Anfragen	Dateien	kb	Besuche	Land
1	1543	99.81%	1271	99.76%	10684	99.85%	231	100.00%	Unbekannte Adressen

Program

```
import socket
import struct
import time
import os
def checksum(data):
    total=0
    length=len(data)
    for i in range(0,length-1,2):
        total += data[i]<<8) + data[i+1]
    if length%2:
        total += data[-1]<<8
    return ~(total)&0xffff
#main program
    host = input()
    for i in range(4):
        ping(host)
        time.sleep(1)
```

Result:

Thus a ping program was successfully implemented using python sends echo request to the specified host

Sample output

from 8.8.8.8 bytes = 48 times = 242

= 2387

= 24.12 ms

= 24.36 ms

Monthly statistics

March 2004 | Last 12 Months | Monthly Statistics | Daily Statistics | Hourly Statistics | URLs | Entry Pages | Exit Pages | Errors | Hosts | Referrer | Searches | Usernames | Useragents | Countries

Tages-Statistik im Monat Marz 2004

Seiten - Dokument-Abfragen (z.B. html, asp, txt) Dateien - erfolgreiche Anfragen für alle Typen (z.B. html, gif, css). Treffer - alle Anfragen, inklusive Fehler.

Tag	Anfragen			Dateien			Seiten			Besuche			Rechner			kb	
	Summe	Schnitt	Maximum	Summe	Schnitt	Maximum	Summe	Schnitt	Maximum	Summe	Schnitt	Maximum	Summe	Schnitt	Maximum	Summe	Schnitt Maximum
7	197	12.74%	25	29		147	11.54%	18	21		186	18.94%	23	27		15	6.41% 2	3		14	8.19% 2	3		2383	22.27%	209	542
8	559	36.16%	24	63		478	37.52%	21	56		419	42.67%	18	40		57	24.36% 2	7		48	28.07% 2	7		4509	42.14%	186	385
9	232	15.01%	11	29		205	16.09%	10	29		118	12.02%	6	16		49	20.94% 2	5		44	25.73% 2	5		955	8.93%	41	138
10	286	18.50%	14	102		205	16.09%	10	34		124	12.63%	6	28		46	19.66% 2	5		38	22.22% 2	4		1336	12.49%	67	379